Data Viz Assignment

COMM 335 - Section 921
Group 2

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Research Question

Which Formula 1 constructor team has demonstrated the best overall performance over the past five years (2020-2024)?

Data Sources and Collection

Our analysis utilized public Formula 1 race data spanning the 2020-2024 seasons, sourced from official FIA (International Automobile Federation) race results and timing databases. The dataset includes race finishing positions, championship points, race wins, and pit stop performance metrics across all Grand Prix events during this period. These datasets were obtained from Kaggle and GitHub and synthesized as per the research question. The sources can be found here.

Key Performance Metrics

We evaluated constructor performance using the following primary indicators:

- **Average Finishing Position**: *Lower values indicate better consistent performance*
- **Total Championship Points**: *Cumulative scoring as achieved by both drivers of the constructor (team)*
- **Race Wins**: *Total victories achieved by each constructor (team)*
- **Pit Stop Efficiency**: *Average pit stop duration, with lower values as a measure of operational excellence*
- **Lap times**: *Fastest average lap sorted by team, with lower values indicating better team performance*

Data Transformations and Analysis

Our team chose the dataset (as linked above) due to the volume and variety of data it provides. Evaluating the performance of constructors is multi-faceted due to the different criteria owing the functions of teams in F1, including but not limited to circuits, constructor details, drivers in each team, lap and pit stop times, qualifying and race results, and overall performance through the course of each season. Through this, we analyzed the 14 teams that participated in the 107 races held over the past 5 seasons.

- Our first visualization is a map locating all circuits and the teams that have won there over the seasons. This can indicate the adaptability and strategic robustness of each team since each race is conducted under different conditions due to track layout and weather.
- Generally, the number of wins correlates with who the champion constructor for each season is. While standings are determined by points achieved and not wins, winning a race brings in a significant haul of points for the team.
- We have also included a points tally visualization since, ultimately, that is what determines the overall performance of teams over the seasons.
- The fastest average lap visualization represents a culmination of driver skill as well as team strategy and car quality (speed and technical functioning).
- Similarly, the relationship between the average starting position and total wins are a holistic representation of team performance. The qualifying sessions during each grand prix determine the starting position for each driver and are fought for through a joint effort of the driver skills (who aim to drive as fast as possible) and constructors (who aim to optimize the car setup). The relationship is similar during the race with the added effect of strategies by the team to maintain or gain position, ultimately securing the win for the best combination.

To achieve these visualizations, we first cleaned up and organized the data by first creating new columns for unique data points (like fastest lap and pit stop times). We also numericalized “null” or “\N” columns by replacing them with “0” to perform all calculations such as sums, averages, and counts. We also manipulated the "URL" column for the constructors dataset in order to pull images of each car from an online image address. This required a formula since it was a “conditional column”.

Dashboard Image